

01 · Who should we email? — customer-level uplift [ANCHOR]

ANCHOR A

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The business decision. We have a €10 discount and a customer list. Emailing *everyone* burns margin on people who would have bought anyway, and a discount email can even annoy some loyal customers into unsubscribing. Emailing *no one* leaves money on the table. So the question is not “does the email work on average?” but “**for which individual customers does sending the email actually change what they do?**”

1.0.1 The one idea this whole notebook rests on

For each customer there are two possible futures: the spend we’d see **if we email them**, and the spend we’d see **if we don’t**. Statisticians call these two futures *potential outcomes* and write them $Y(1)$ (emailed) and $Y(0)$ (not emailed). The thing we care about is the **difference** — how much the email *changes* that customer’s spend:

$$\tau_i = Y_i(1) - Y_i(0). \quad (\text{the customer’s individual treatment effect, in €})$$

We use the Greek letter **tau** (“**tau**”) for this effect throughout. When it depends on the customer’s characteristics x (recency, engagement, ...) we write it as a function $\tau(x)$.

The catch — the “**fundamental problem of causal inference**” — is that for any one customer we only ever get to see *one* of the two futures: we either emailed them or we didn’t, never both. So τ_i is never directly observed for anybody. Everything that follows is a disciplined way of *filling in the future we didn’t see*.

1.0.2 Four kinds of customer — and why only the effect tells them apart

Sort customers by what they do in each of the two futures, and four types appear. Notice that two customers can have the *same observed behaviour* (“bought after being emailed”) yet be completely different types:

	buys if emailed	doesn't buy if emailed
buys if not emailed	(yellow) <i>Sure Thing</i> — would buy anyway, so $\tau \sim 0$ (the discount is pure waste)	(red) <i>Sleeping Dog</i> — the email backfires , $\tau < 0$ (reminding them prompts an unsubscribe/return)
doesn't buy if not emailed	(green) <i>Persuable</i> — the email <i>tips them over</i> , $\tau > 0$ (the only type worth paying to reach)	(white) <i>Lost Cause</i> — won't buy regardless, $\tau \sim 0$

Only the **Persuadables** are worth a discount. But you cannot identify them from who bought — a *Sure Thing* and a *Persuable* both “bought after the email.” You can only separate them by estimating the *effect* τ , which is why plain response models (predicting who buys) are the wrong tool and **uplift models** (predicting whose behaviour *changes*) are the right one.

Targeting is worthwhile **only because tau varies across customers** (“heterogeneity”). If every customer had the same τ , there'd be nothing to select on — you'd rationally email everyone (if $\tau > \text{cost}$) or no one (if $\tau < \text{cost}$). So the entire problem reduces to: **estimate the per-customer effect $\tau(x)$ well, then turn it into a euro decision that respects how unsure we are.**

1.0.3 How this notebook is organised

It follows the repo's fixed **7-step contract**, then adds **three depths** a data scientist needs to defend the number to a skeptic:

- **Steps** — 1 question · 2 simulate a known truth · 3 identify · 4 estimate · 5 validate · 6 decide in € · 7 caveats
- **Depth A** estimator bake-off & failure modes · **Depth B** identification rigour & sensitivity · **Depth C** euro policy, value-of-information & test design

A note on “Bayesian.” We fit models that return not a single number but a *distribution* of plausible values for each effect — a **posterior**. In the narration we never dwell on the machinery; a posterior just lets us say “*how sure are we, and does that change what we'd do?*” A **90% credible interval** below means “given the data and model, there's a 90% chance the true value lies in this range.”

```
FAST=True N=900 sampling={'draws': 140, 'tune': 140, 'chains': 2, 'm': 30}
```

1.1 2 · Simulate a ground truth (why we start with fake data)

Here is the uncomfortable fact that shapes how we *validate* any causal method: **on real data we can never check whether the method got the effect right**, because the true effect τ_i is never observed for anyone (we only saw one of their two futures). If we just ran a fancy model on real data and it printed “€6.10”, we'd have no way to know if that's right or nonsense.

The way out — used in every notebook here — is **validation-first**: we build a *simulator* that plants a **known** effect for each customer, generate data from it, and then see whether the method **recovers the number we planted**. Only after a method passes that test on data where we know the answer do we trust it on data where we don't. (Section 6b then runs the same pipeline on a real public dataset, where we can no longer grade recovery but can still make the decision.)

What the simulator plants. Each customer gets:

- five features x — **recency** (days since last purchase), **frequency**, **monetary**, **tenure**, **engage** (an engagement score);
- a true per-customer effect $\tau(x)$ (in €), deliberately *heterogeneous* — engaged, mid-recency customers are **Persuadables** (large positive tau); habitual high-frequency but low-engagement customers are **Sleeping Dogs** (tau < 0, the email nudges them to unsubscribe); the rest sit near 0;
- a **selection rule** that decides who got emailed in the past. In the *observational* world we use here, marketers historically emailed the already-engaged, recently-active customers — so whether a customer was emailed is **tangled up with the kind of customer they are**. That tangling is called **confounding**, and it is exactly what makes the naive answer wrong (Step 4 and Depth A show it).

True ATE (average of tau over everyone) = €6.07

Share historically emailed = 47%

Share of each uplift type in the base:

sure thing / lost cause 0.39

persuadable 0.32

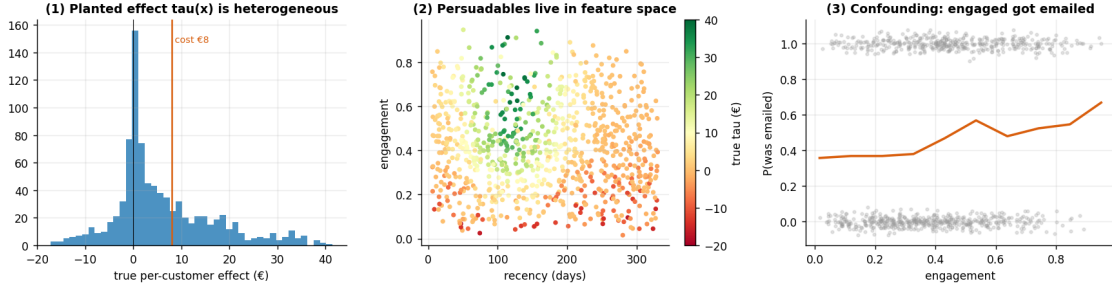
sleeping dog 0.29

	recency	frequency	monetary	tenure	engage	T	y \
0	208.156027	6.0	108.310843	13.737214	0.276899	0.0	42.175390
1	296.594485	3.0	142.587867	25.200485	0.473980	0.0	67.436225
2	257.097849	4.0	107.168786	57.141597	0.390539	0.0	48.743181
3	78.192337	7.0	98.732666	13.421173	0.086593	0.0	46.782878
4	102.554043	7.0	87.199065	4.999605	0.359222	1.0	69.098355

	tau
0	-0.457966
1	-0.028135
2	0.129894
3	-9.869026
4	21.912770

The three panels below are the picture to hold in your head for the rest of the notebook:

1. **The planted effect is spread out** (not a single value) — that spread is the *heterogeneity* that makes targeting possible, and it straddles both 0 and the €8 cost line.
2. **Persuadables cluster in feature space** — high tau (green) lives at particular recency/engagement combinations, which is *why* a model using x can find them.
3. **The confounding, made visible** — the probability of having been emailed *rises with engagement*. Emailed and not-emailed customers are therefore **not comparable as-is**; that is the problem every method below has to overcome.



1.2 3 · Identify — what exactly are we estimating, and when is it even possible?

“Identification” vs “estimation” — the single most important distinction in causal inference. *Identification* asks: *given how the world works, can the effect in principle be recovered from this data at all?* That is a question about **assumptions**, answered before you fit anything. *Estimation* is the statistics you do *once identification is granted*. A beautiful model computed under a false identification assumption is **confidently wrong** — so we keep the two separate.

The estimand (the precise thing we want). We climb a ladder of three related quantities:

quantity	plain meaning	formula
ATE — Average Treatment Effect	“does emailing lift spend <i>overall?</i> ”	$\mathbb{E}[Y(1) - Y(0)]$
CATE — Conditional ATE, $\tau(x)$	“the effect <i>for customers like x</i> ” -> <i>who</i> to target	$\mathbb{E}[Y(1) - Y(0) \mid X = x]$
policy value	“the € we make from a given targeting rule”	$\mathbb{E}[(\tau(x) - c) \mathbb{1}\{\text{we target } x\}]$

($\mathbb{E}[\cdot]$ is just “average over customers”; $\mathbb{1}\{\cdot\}$ is 1 when the condition holds, else 0.) The uplift decision needs all three, but everything rests on the CATE $\tau(x) = \mu_1(x) - \mu_0(x)$, where $\mu_t(x) = \mathbb{E}[Y \mid T = t, X = x]$ is the *average spend of emailed (t=1) or not-emailed (t=0) customers who look like x*.

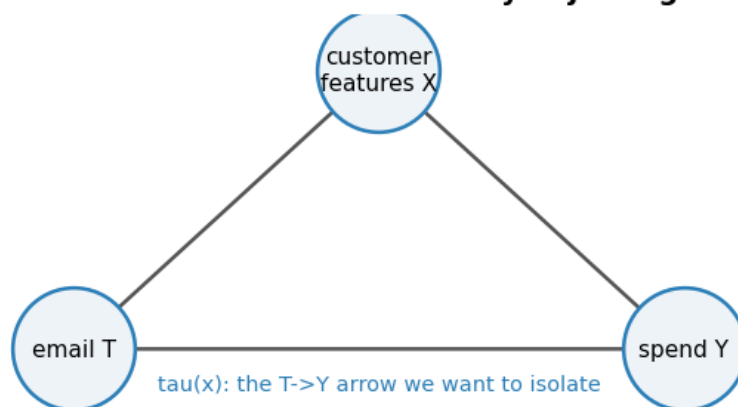
The three assumptions that make $\tau(x)$ recoverable (each is revisited in Depth B):

1. **Unconfoundedness** (a.k.a. *no unmeasured confounders*, “ignorability”): once we condition on the measured features X , who got emailed is as-good-as-random. Formally $\{Y(0), Y(1)\} \perp T \mid X$. *Guaranteed by design in an A/B test; an assumption otherwise* — and the big one, because it can’t be checked from data.
2. **Positivity / overlap**: every kind of customer had *some* chance of being emailed and *some* chance of not — $0 < e(x) < 1$, where $e(x) = P(T=1 \mid X=x)$ is the **propensity score** (a customer’s probability of having been emailed). If some group was *always* emailed, we have no comparison for them and are extrapolating. Checked in Step 4.
3. **SUTVA**: one customer’s email doesn’t change another’s outcome (no word-of-mouth spillover), and there’s only one version of “the email.”

Under 1–3, the **backdoor adjustment** (or *g-formula*) says $\tau(x) = \mu_1(x) - \mu_0(x)$: comparing emailed vs not-emailed *within* look-alike customers removes the confounding. The DAG (directed

acyclic graph — a picture of what causes what) shows why. The “backdoor path” $\text{email} \leftarrow \text{features} \rightarrow \text{spend}$ is the route by which confounding sneaks in; conditioning on the features **blocks** it.

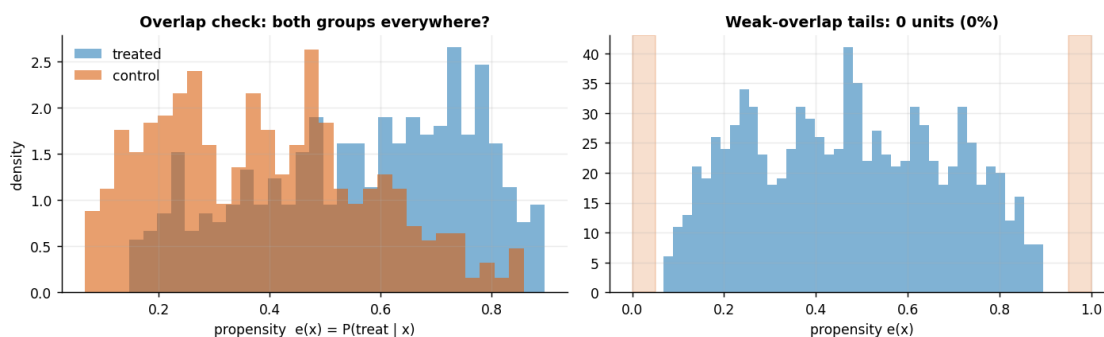
Backdoor $T \leftarrow X \rightarrow Y$ is closed by adjusting for X



1.3 4 · Estimate — check overlap first, then fit two independent ways

Discipline: look at overlap *before* estimating. Adjustment can only compare emailed vs not-emailed customers where *both* actually exist. If, at some feature combination, everyone was emailed, there is no control to compare against and the model is silently extrapolating. So we first plot the **propensity score** $e(x)$ by arm (emailed vs not); healthy overlap means the two histograms cover the same range, with few customers stuck at 0 or 1.

Overlap is adequate - only 0 of 900 customers fall in the <0.05 / >0.95 tails.



Now the estimators. We fit the effect two completely different ways and check they agree — agreement across independent methods is our first credibility signal.

- **BCF — Bayesian Causal Forest** (our primary CATE model). Under the hood it’s **BART** (Bayesian Additive Regression Trees — a flexible sum-of-trees model that needs no functional-

form guesswork and returns a posterior). BCF is a causal-tuned BART with two tricks: it feeds the estimated propensity $e(x)$ into the model so *targeted-selection confounding* is absorbed, and it puts a **tighter prior** on the treatment part than the baseline part, so the model only reports heterogeneity when the data really support it (rather than reading noise as signal). Output: a posterior array of shape (draws, customers) — one plausible $\tau(x)$ curve per posterior draw.

- **AIPW — Augmented Inverse-Propensity Weighting** (a cross-check on the *average*). It's **doubly-robust**: it combines an outcome model and a propensity model, and is correct if *either one* is right. Because it leans on entirely different machinery than BART, matching ATEs is reassuring.

The pymc-bart gotcha, baked in. Scoring BART counterfactuals (“what would this customer have spent under the *other* treatment?”) silently returns *frozen training predictions* — every effect exactly 0 — unless the tree node is resampled. `cmp.estimators` always does this correctly, and a unit test asserts non-zero recovered effects, so you never hit the trap.

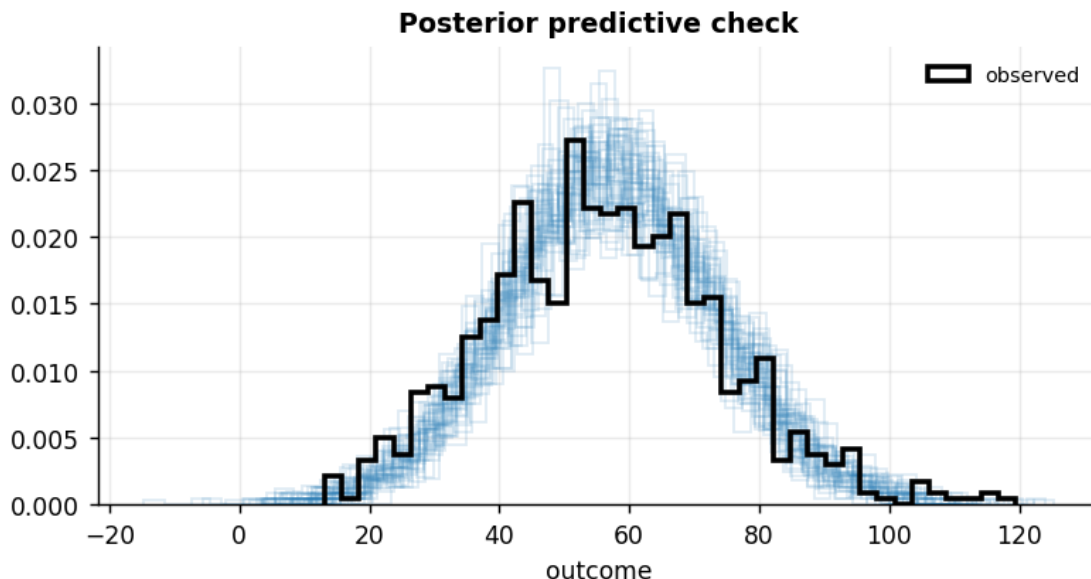
BCF ATE = €6.57 (true €6.07)

AIPW ATE = €5.85 90% CI [5.19, 6.99] (doubly-robust)

Two independent estimators landing on the same ATE is our first credibility check.

Model criticism — the posterior predictive check (PPC). Before we *read effects off* a model, we ask whether it can even reproduce the data it was trained on. We draw fake (“replicated”) datasets from the fitted model and overlay their spend distribution on the real one. If the real data looked nothing like what the model can generate (wrong spread, missed skew), the effect estimates would be untrustworthy no matter how tight their intervals.

Observed spend (black) sits inside the replicate envelope (blue) - no gross misfit (residual sigma ~ €15).



1.4 5 · Validate — did we recover the truth, is the uncertainty honest, and does it rank well?

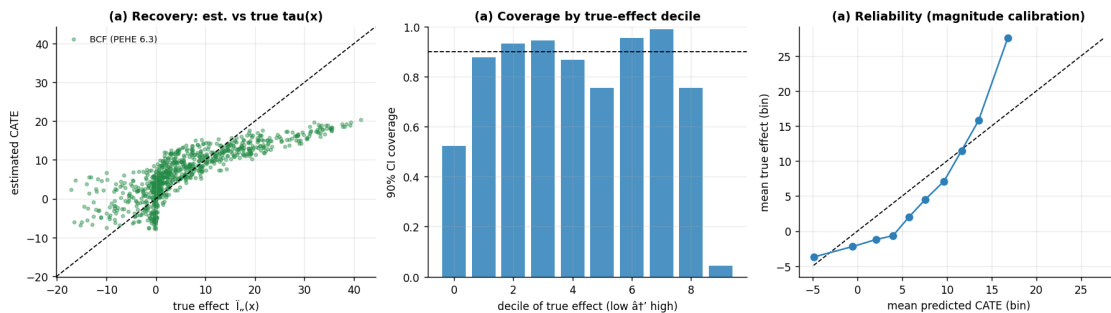
This is the payoff of simulating a known truth. Three questions, each with its own diagnostic:

(a) **Recovery & calibration** — are the numbers right, and honestly uncertain? - **PEHE** (*Precision in Estimation of Heterogeneous Effects*) — the root-mean-square error between the estimated $\hat{\tau}(x)$ and the true $\tau(x)$, in €. Lower is better; it's the CATE analogue of RMSE. - **Coverage** — a 90% *credible interval* should contain the true effect about 90% of the time. We also break coverage out *by decile of the true effect* to catch a model that's well-calibrated on average but overconfident for the biggest effects. - **Reliability** — bucket customers by predicted effect and plot mean-predicted vs mean-true; on the 45° line means the *magnitudes* are calibrated, not just the ranking. - **Sharpness** — the average width of the intervals. Sharp *and* calibrated is the goal; sharp *and* miscalibrated is just overconfidence, which is why we never report sharpness alone.

(b) **Ranking quality** — for targeting, the *order* matters more than exact magnitudes. - **Qini / uplift curve** — walk down customers from highest to lowest predicted effect, contacting as you go, and plot the cumulative *true* effect captured. A good model's curve bows up toward the **oracle** (a hypothetical model that knows every true tau) and well above the **random** diagonal. - **AUUC** (*Area Under the Uplift Curve*) — that curve's area, normalised so **1.0 = oracle-perfect ranking** and **0 = no better than random**. The single number for “how well do we rank?” - **Uplift-by-decile** — average *true* effect within each tenth of predicted effect; a good model shows a clean descending staircase (top decile = biggest real effect).

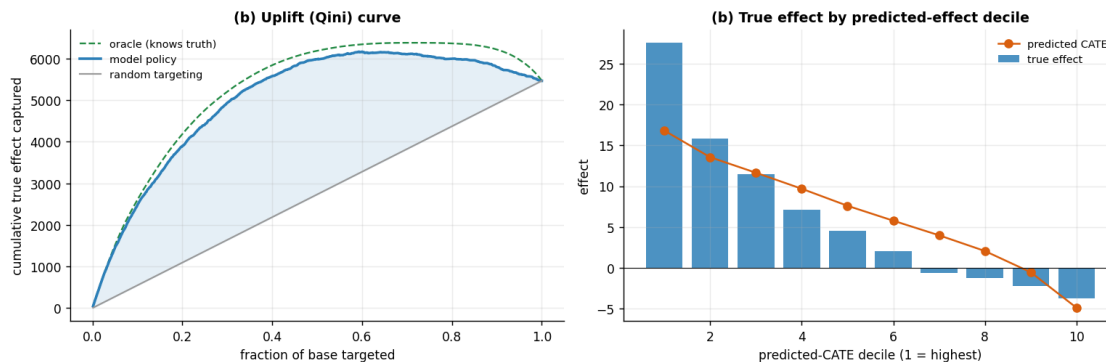
(c) **Stability** — is the result a fluke of one random draw? We refit on several fresh simulations and check the recovery is consistent.

PEHE €6.34 · corr(est,true) 0.83 · 90% coverage 76% · interval sharpness €13.6



AUUC = 0.90 of oracle (1.0 = perfect ranking, 0 = random).

Top decile true effect €27.6 vs bottom decile €-3.7 - the model concentrates the real uplift where we'd actually spend.



(c) Stability across fresh simulations:

seed	ATE_hat	true_ATE	PEHE	AUUC
100	6.72	5.74	6.49	0.90
101	4.72	5.83	6.24	0.91
102	6.03	5.44	5.77	0.91
103	5.84	5.20	6.26	0.89

ATE recovery bias across seeds: +0.27 (sd 0.94) - consistent, not a lucky seed.

1.5 6 · Decide, in euros

An effect estimate is useless until it becomes an action and a number a manager signs off on. Three moves:

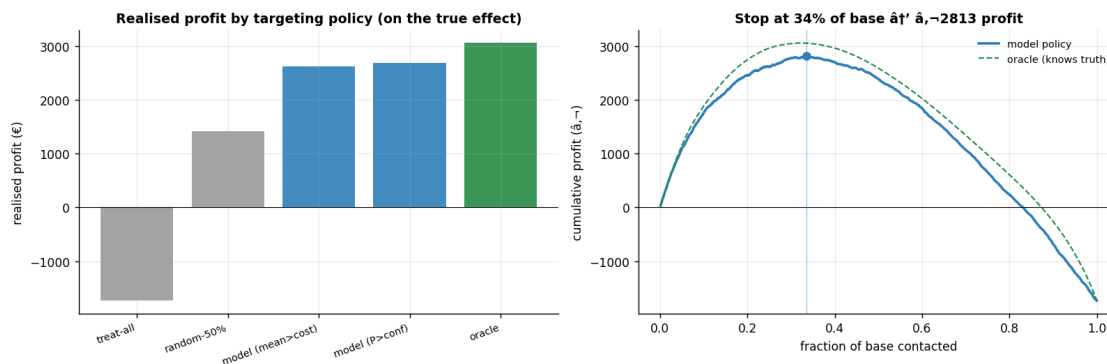
Compare targeting policies on realised profit. We evaluate five rules on the *known truth* (only possible in simulation) so we can see, in €, how much targeting beats the alternatives and how close it gets to the unbeatable oracle: - *treat-all* (email everyone), *treat-none* (baseline), *random-50%*, - *model* (*mean* > *cost*) — email where the posterior-mean effect beats the €8 cost, - *model* (*P* > *conf*) — the **honest rule**: email only where we’re *confident* the effect beats cost (see below), and - *oracle* — email exactly the customers whose *true* effect beats cost (the best any model could do).

The honest targeting rule. We do **not** target on “mean effect > cost.” A customer whose point estimate says “target” but whose interval straddles the cost line is a coin-flip, not a yes. So we target where **\$P((x) > cost) \$ is high** (here > 0.8) — probability, from the posterior, that the effect really clears the cost.

The profit curve. Rank customers by estimated effect, contact down the list, and plot cumulative (true effect - cost). It rises while we’re adding profitable customers and falls once we hit unprofitable ones; its **peak is the optimal stopping point** — “email the top X%, make €Y.”

policy	profit	frac_contacted	profit_vs_all
treat-all	-1734.0	1.0	0.0
random-50%	1414.0	1.0	3148.0
model (mean>cost)	2620.0	0.0	4355.0
model (P>conf)	2681.0	0.0	4416.0
oracle	3058.0	0.0	4792.0

Email everyone: €-1,734 · honest model rule: €2,681 (88% of the oracle's profit), targeting 236 of 900 customers.



1.6 6b · The same pipeline on real data (Hillstrom email experiment)

Everything above used *simulated* data so we could grade recovery. Here we run the identical estimator on a **real, public dataset you don't have to provide** — it's fetched over the network from a public URL:

Hillstrom / MineThatData (2008) — 64,000 real customers randomly assigned to *Men's email*, *Women's email*, or *no email*, with downstream visits, conversion and spend. We reduce it to a binary *Women's-email vs no-email* comparison. Because assignment was **randomized (a true A/B test)**, the average effect is trustworthy with no adjustment — but there is **no ground-truth per-customer effect** here (real customers only live one future), so we can estimate and decide, but we *cannot* grade CATE recovery. That's exactly what production looks like: the validation lives in the simulation above; the money is made here.

The cell is **gated** — it fetches only if you set `CMP_REAL=1` (so the offline test suite stays deterministic). Set it and re-run to see the real-data version.

Real-data section skipped. Set `CMP_REAL=1` and re-run this cell to fetch the Hillstrom experiment.

On real data there is no ground-truth tau, so Steps 5's recovery checks do not apply.

1.7 7 · Caveats — the honest failure modes

- **Unconfoundedness is untestable.** If a *hidden* driver moved both who got emailed and how much they spend (say a salesperson's hunch we never recorded), the estimate drifts. Depth B quantifies exactly how strong such a confounder would have to be to flip the decision — as an **E-value**.
- **Overlap matters.** Where some kind of customer was *always* emailed, we're extrapolating, not comparing. We checked the tails in Step 4 and can trim them.

- **Estimator choice is not cosmetic.** The S-learner (Depth A) regularises the treatment signal away and under-detects heterogeneity — it fails on both PEHE *and* ranking (AUUC).
- **BCF ~ T-learner when confounders are fully observed.** The propensity trick earns its keep specifically under *targeted selection*; don't oversell it as free lunch.
- **Sleeping dogs are real.** Negative-effect customers mean “email everyone” can be *value-destroying*, not merely wasteful — which is exactly what the policy comparison in Step 6 showed.

2 Depth A · Estimator bake-off & failure modes

Why a whole section on picking the estimator? Because the choice is not cosmetic — a popular default (the S-learner) systematically **flattens** heterogeneity and would quietly wreck the targeting. We line up three estimators against the known truth, in two worlds, on the metrics that actually matter for targeting.

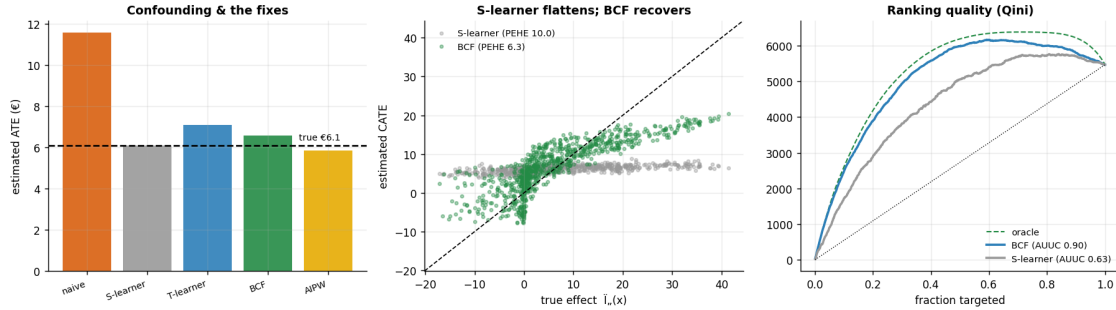
The three estimators (meta-learners — recipes that turn any regression into a CATE estimator): - **S-learner** (“S” = single) — fit *one* model of spend on features *and* treatment together, then read the effect as (prediction with $T=1$) — (prediction with $T=0$). Simple, but the model can shrink the treatment's influence toward zero, **flattening** real differences. This is the labelled failure mode. - **T-learner** (“T” = two) — fit *separate* models on the emailed and not-emailed customers, then subtract. A solid default when treatment is randomized. - **BCF** — the propensity-aware, shrink-to-homogeneity model from Step 4; best under confounding.

The two worlds: - **Randomized** — treatment was a coin flip, so *unconfoundedness holds by design* and even simple methods are unbiased on the average. - **Observational** — the confounded world we've been in; here the naive “emailed — not-emailed” average is badly biased, and the estimator has to actively undo the confounding.

The metrics: PEHE (magnitude error, €), corr (ranking correlation), **AUUC** (ranking quality vs oracle), ATE_bias (average-effect error, €), and cov90 (are the 90% intervals honest?).

Naive 'emailed - not emailed' (observational): €11.6 vs true €6.1 -> bias +5.5
(confounding nearly doubles the apparent effect)

	regime	estimator	PEHE	corr	AUUC	ATE_bias	cov90
0	randomized	S-learner	10.53	0.54	0.60	-3.20	0.72
1	randomized	T-learner	6.42	0.80	0.86	0.13	0.86
2	randomized	BCF	7.12	0.77	0.83	-0.11	0.82
3	observational	S-learner	10.00	0.58	0.63	0.01	0.73
4	observational	T-learner	5.85	0.84	0.90	1.00	0.93
5	observational	BCF	6.34	0.83	0.90	0.49	0.76



Read-out. In the randomized world all three are roughly unbiased on the ATE, but the S-learner already ranks worse (lower AUUC, higher PEHE) — flattening shows up even without confounding. In the observational world the **naive average is badly biased**; the S-learner both misses the heterogeneity *and* is mis-calibrated; **BCF gets the closest per-customer recovery, the best ranking, and near-nominal coverage** — and it agrees with the completely separate **AIPW estimator on the ATE**. Correct ranking *plus* honest uncertainty is exactly what the euro decision needs.

3 Depth B · Identification rigour & sensitivity

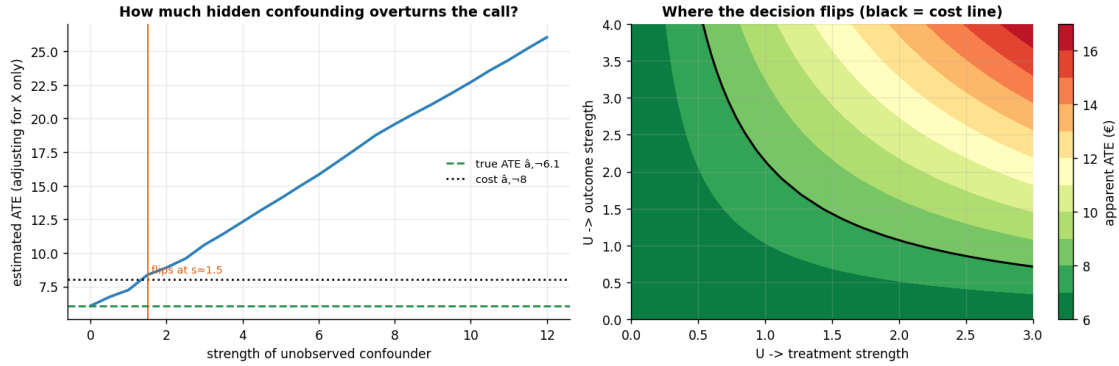
We already confirmed **overlap** (Step 4). The deeper worry is **unconfoundedness**, which is *untestable* — so the honest question is not “is it true?” but “**how fragile is our conclusion to a confounder we didn’t measure?**”

We simulate a hidden driver U that moves *both* who gets emailed and how much they spend, sweep its strength, and refit adjusting for the observed features **only** (blind to U , as we’d be in reality). We report two things a CMO can act on: - the **tipping point** — the confounder strength at which the (wrongly) adjusted effect crosses the €8 cost line and would flip us from “target selectively” to “email everyone pays”; - the **E-value** (VanderWeele) — the minimum strength of association a hidden confounder would need with *both* treatment and outcome, on a risk-ratio scale, to explain the effect away entirely. A large E-value means “you’d need an implausibly strong hidden confounder to overturn this,” i.e. the finding is robust.

Tipping point: confounder strength ~ 1.5 . E-value ~ 5.76 - a hidden confounder would need

about a 5.8x association with BOTH email and spend (risk-ratio scale) to explain the effect

down to break-even. Ask the domain expert whether anything that strong plausibly went unmeasured.

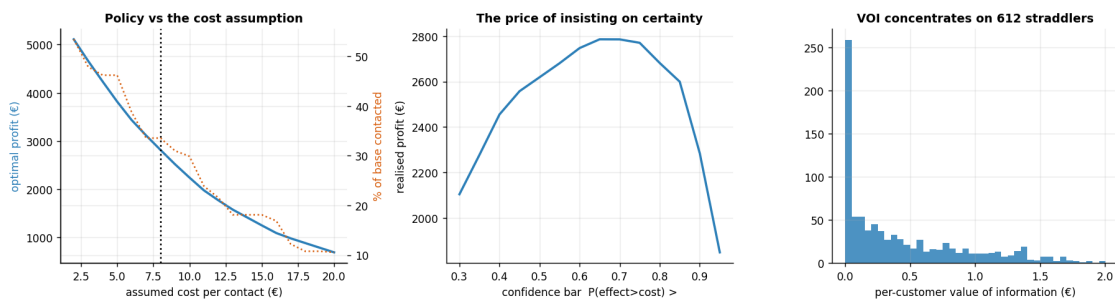


4 Depth C · Euro policy — sweeps, value of information & test design

The decision is a number *under assumptions*. We stress the two a manager will argue about — the per-contact **cost** and how **confident** we insist on being — and then price the remaining uncertainty and turn it into a concrete experiment.

Value of information (VOI). Uncertainty has a *price*: the euros we lose by acting on the noisy estimate instead of the (unknown) truth. It concentrates on the **straddlers** — customers whose credible interval crosses the cost line, so we genuinely can't tell if they're worth contacting. The total VOI on the straddlers is, quite literally, **the maximum you'd rationally pay for a clean experiment on them** — which tells you how big an A/B test to run, and on whom.

Total value of information €388; €383 of it sits on the 612 'straddler' customers (68% of the base) whose interval crosses the cost line.



4.0.1 The one-paragraph decision

Targeting the model's *confident* persuadables makes materially more money than emailing everyone (which here **loses** money to sleeping dogs), and captures most of the oracle's profit. The call is robust to plausible cost assumptions and would only reverse

under fairly strong *hidden* confounding (Depth B's E-value). The uncertainty that remains is worth real money and is concentrated on the “straddler” customers whose interval crosses the cost line — **so the follow-up is a small randomized A/B test on the straddlers, not a blanket send.** Act where we're sure; measure where we're not. (*In FAST mode the posterior is deliberately coarse, so the straddler set is wide; a full-quality run tightens the intervals and shrinks it.*)

```
{
  "true_ate": 6.0732099315101244,
  "bcf_ate": 6.565869203310453,
  "aipw_ate": 5.851767331319051,
  "PEHE": 6.337910340227753,
  "AUUC": 0.8996165493371009,
  "coverage90": 0.7644444444444445,
  "profit_treat_all": -1734.1110616408864,
  "profit_model": 2681.463055702852,
  "profit_oracle": 3057.9390084417983,
  "sensitivity_tip": 1.5,
  "e_value": 5.756405386243919,
  "VOI_total": 388.0348053948478,
  "n_straddlers_to_test": 612
}
```